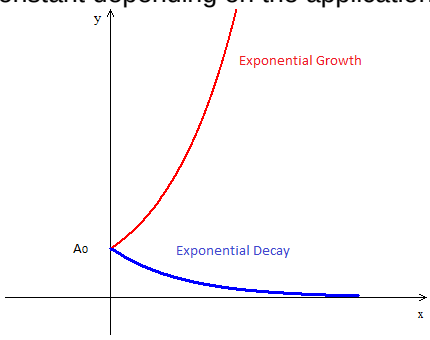
8.7 Solving Problems with Exponential and Logarithmic Functions

A Exponential Growth and Decay

Exponential Growth and Decay may be modelled by a function of the form:

$$A(t) = A_0(b^{kt})$$

where
t is time
A_0 is the initial amount
$A(t)$ is the amount at time t
b is the base
k is a constant depending on the application



Ex 1. Let $f(x) = 4(2^{4x-1})$. **Skip This**

a) Write this relation in the form $f(x) = Ab^{Bx}$.

$f(x) = 4 2^{4x} \cdot 2^{-1} = 2 \cdot 2^{4x}$

b) Write this relation in the form $f(x) = Ab^{x}$.

$f(x) = 2(16^{x})$

c) Write this relation in the form $f(x) = A(10^{Bx})$.

$f(x) = 2\, 10^{x \log_{10} 16}$

d) Write this relation in the form $f(x) = A(3^{Bx})$.

$f(x) = 2 \cdot 3^{x \log_3 16}$

B Common Ratio **Skip This**

The values of the exponential growth function form a geometric sequence:

$$\frac{y_2}{y_1} = \frac{y_3}{y_2} = \ldots = \frac{y_{n+1}}{y_n}$$

where

$y_1 = f(x_1)$, $y_2 = f(x_2)$, …

and x_1, x_2, … are in arithmetic sequence.

Ex 2. Show that the following relation is exponential.

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$f(x) = 2 \cdot 3^{(x-1)}$

x	y	
1	2	
2	6	3
3	18	3
4	54	3
5	162	3

C Developing Exponential Growth Formula

If r is the increasing rate per year, use:

$$A(t) = A_0(1+r)^t$$

Indeed $A(1) = A_0(1+r)^1 = A_0 + rA_0$.

If, over a period T, the amount is increasing b times, use:

$$A(t) = A_0 b^{\frac{t}{T}}$$

Indeed $A(T) = A_0 b^{\frac{T}{T}} = bA_0$.

Ex 3. For each case, find an exponential function that model best the situation.

a) The value of a house is increasing by 7% per year.

$V = V_0(1+0.07)^t = V_0(1.07)$

b) The number of bacteria is triple every two hours.

$N = N_0 3^{t/2}$

c) The number of bacteria is double every five hours.

$N = N_0 2^{t/5}$

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D Developing Exponential Decay Formula Exponential Decay may be modelled by a function of the form: $$A(t) = A_0(b^{kt})$$ or by $$A(t) = A_0\left(\frac{1}{2}\right)^{\frac{t}{H}}$$ where H is half-life $$A(H) = \frac{A_0}{2}$$ or by $$A(t) = A_0(1-r)^t$$ where r is the decreasing rate per year $$A(1) = A_0(1-r)^1 = A_0 - rA_0$$	Ex 4. For each case, find an exponential function that model best the situation. a) The value of a car is decreasing by 5% per year. $V = V_0(1-0.05)^t$ $= V_0(0.95)^t$ b) The half-life of a radioactive source is 81 years. $m = m_0\left(\frac{1}{2}\right)^{t/81}$ c) The luminosity decreases 3 times for each 10cm of depth. $I = I_0\left(\frac{1}{3}\right)^{\frac{x}{10}}$
E Sound Level $$L = 10\log\left(\frac{I}{I_0}\right)$$ where L is the soundness (sound level) in decibels I is the intensity of the sound $I_0 = 10^{-12} W/m^2$ is a constant (intensity of the sound at the threshold of hearing) Note. $L_2 - L_1 = 10\log(I_2/I_1)$	Ex 5. A whisper has a sound level of 15 dB and a rock concert has a sound level of 120 dB. How many more intense is the rock concert in comparison to a whisper? $L_1 = 10\log\frac{I_1}{I_0}$ $\quad \frac{I_1}{I_0} = 10^{\frac{L_1}{10}}$ $L_2 = 10\log\frac{I_2}{I_0}$ $\quad \frac{I_2}{I_0} = 10^{\frac{L_2}{10}}$ $\frac{I_2}{I_1} = 10^{\frac{L_2-L_1}{10}} = 10^{\frac{120-15}{10}} = 10^{10.5}$ times
F Earthquake Magnitude $$M = \log\left(\frac{A}{A_0}\right)$$ where M is the magnitude of the earthquake A is the amplitude (intensity) of the earthquake A_0 is a constant Note. $M_2 - M_1 = \log(A_2/A_1)$	Ex 6. In 2017, in Mexico, two earthquakes happened with a magnitude more than 7. One happened on September 7 and had a magnitude of 8.2 and the other happened on September 19 and had a magnitude of 7.1. How many times was the amplitude of the September 7 earthquake greater in comparison to the amplitude of the September 19 earthquake? $M_1 = 8.2$, $M_2 = 7.1$ $\quad M = \log\frac{A}{A_0}$ $\frac{A_1}{A_2} = 10^{M_2-M_1} = 10^{8.2-7.1} = 10^{0.9}$
G pH Scale $$pH = -\log n$$ where pH is a number measuring acidity/alkalinity of a substance $n = [H^+]$ is the concentration of hydrogen ions	Ex 7. Lemon juice has a pH of 2.5 and milk has a pH of 9. How many times the hydrogen ions are more concentrated in lemon juice than in milk. $pH_1 = 2.5$ $pH_2 = 9$ $\frac{[H^+]_1}{[H^+]_2} = \frac{10^{-2.5}}{10^{-9}} = 10^{9-2.5}$ $= 10^{6.5}$ times.

Reading: Nelson Textbook, Pages 493-499
Homework: Nelson Textbook, Page 499 #1-5, 8, 10, 14, 15, 17, 18

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